

Tutorial-4.

Logic Regression.

An email service Provider wants to classify emails as spam & Not spam using logistic regression. Data from 10 emails has been collected.

Input Features:

- Number of suspicious words
- Number of External Links
- Email Length (words)
- Presence of Attachments (0 = No, 1 = Yes)

Target Variable:

- Spam (1 = Spam, 0 = Not Spam)

suspicious
(x_1) words

External
(x_2) links

Attachment
(x_3)

Spam
(y)

5

2

1

0

3

1

0

0

↓

↓
 w_1

↓
 w_3

Bias: $w_0 = 0$

$$Z = w_0 + w_1x_1 + w_2x_2 + w_3x_3$$

$$Z_1 = 0 + 1(5) + 1(2) + 1(1)$$

$$= 5 + 2 + 1$$

$$= 8$$

$$Z_2 = 0 + 1(1) + 1(0) + 1(0)$$

$$= 1$$

$$Z_3 = 4$$

$$Z_4 = 0$$

$$\hat{y}_{\text{pred}} = \sigma(z)$$

$$= \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-9}} = 0.99$$

$$\Rightarrow 0.99 \geq 0.5 \Rightarrow \text{so it is 1}$$

$$0.78 \geq 0.5 \Rightarrow \text{so it is 1}$$

$$0.78 \geq 0.5 \Rightarrow 1$$

$$0.5 \geq 0.5 \Rightarrow 1$$

y - y_{pred}

$$0.01$$

$$-0.78$$

$$0.02$$

$$-0.5$$

$$CF = J = \frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

$$\frac{\partial J}{\partial w} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)$$

$$= \frac{1}{4} (0.01 + (-0.78) + 0.02 + (-0.5))$$

$$CF = -0.3125$$

$$w_a^{\text{new}} = w_a^{\text{old}} + \eta \frac{1}{N} \sum (y_i - \hat{y}_i)$$

$$\because \eta = 1$$

$$= 0 + 1 \frac{1}{4} (-0.3125)$$

$$= -0.078$$

→ Bias update.

Weights:

$$w_i^{\text{new}} = w_i^{\text{old}} + \eta \frac{1}{N} \sum (y_i - \hat{y}_i) x_i$$

$$= w_i^{\text{old}} + \eta \left(\frac{\partial J}{\partial w} \right) x_i$$

$$w_1^{\text{new}} = 1 + \frac{1}{4} [(0.01 \times 5) + (-0.78 \times 1) + (0.02 \times 3) + (-0.5 \times 0)]$$

$$= 0.8325.$$

$$w_2^{\text{New}} = 1 + \frac{1}{4} \left[(0.01 \times 2) + (-0.78 \times 0) + (0.02 \times 1) + (-0.5 \times 0) \right]$$

$$= 1.01.$$

$$w_3^{\text{New}} = 1 + \frac{1}{4} \left[(0.01 \times 1) + (-0.78 \times 0) + (0.02 \times 0) + (-0.5 \times 0) \right]$$

$$= 1.0025.$$

$$z_1^{\text{New}} = w_0^{\text{New}} + w_1^{\text{New}} x_1 + w_2^{\text{New}} x_2 + \dots$$

$$= 0.3125 + 0.8325(5) + 1.01(2) + 1.0025(1)$$

$$= 6.8725$$

$$z_2^{\text{New}} = -0.3125 + 0.8325(1) + 1.01(0)$$

$$+ (1.0025)(0)$$

$$= 0.52.$$

$$z_3^{\text{New}} = -0.3125 + 0.8325(3) + 1.01(1) +$$

$$(1.0025)(0)$$

$$= 3.195$$

$$z_4^{\text{New}} = -0.3125 + 0.8325(0) + 1.01(0) + (1.0025)(0)$$

$$= -0.3125$$

$$y^{\wedge} = \frac{1}{1 + e^{-z}}$$

$$y_1^{\wedge} = 0.999 \quad y_1^{\wedge} = 1 \rightarrow \text{correct prediction}$$

$$y_2^{\wedge} = 0.6272 \quad y_2^{\wedge} = 1 \rightarrow \text{NOT correct prediction}$$

$$y_3^{\wedge} = 0.961 \quad y_3^{\wedge} = 1 \rightarrow \text{correct prediction}$$

$$y_4^{\wedge} = 0.4226 \quad y_4^{\wedge} = 0 \rightarrow \text{correct prediction}$$